SPE-227568-MS

Improving Economic Feasibility of Lithium Extraction from Produced Water

M. P. Yutkin, K. M. Kaprielova, and T. W. Patzek, Physical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal, Kingdom of Saudi Arabia

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

This study investigates an ion-exchange-based approach to enhance lithium concentration in produced brine, with application to geothermal, oilfield, and mining waterflooding operations. Coreflood experiments confirm that lithium actively participates in ion exchange, even in carbonate rocks with limited clay content. Lithium adsorption isotherms reveal that a significant amount of lithium potentially remain stored on exchange surfaces over extended waterflooding periods, unless specific desorption criteria are met. This implies that economically meaningful quantities of lithium could still be recovered from wells with long injection history.

Numerical simulations at large scale show that lithium recovery primarily depends on two factors: the contrast in ionic strength between injected and formation brines, and the cation exchange capacity of the rock. Another simulation result highlights the roles of competing ions. For example, elevated salinity of formation brine can reduce lithium storage capacity. Based on systematic evaluation of numerical results with a wide range of input parameters, we provide general guidelines for enhancing lithium extraction. Economic projections suggest that incremented lithium yields may offset costs in well-flooding projects that would otherwise be considered economically unattractive.

Finally, a short-term single-well injection-flowback scenario is explored. It tests the feasibility of early-stage field evaluation and lithium resource discovery. The model shows that an injection pill of 0.1 pore volume allows for detection of ion exchange through total mass balance, in the presence of a conservative tracer and moderate dispersion. Such tests are fast and inexpensive. They could guide the decision making process for lithium extraction and help identify zones with lithium-loaded exchangers for potential use in the future.

Introduction

Lithium has gained a lot of attention over the last two decades because of its proven potential in electrochemical applications and energy storage. Namely, owing to its light weight, small atomic size, and thus fast mobility it is considered a valuable product for batteries in electric vehicles, portable electronics, and grid storage (Scrosati and Garche 2010; Tarascon and Armand 2001; Goodenough and Park 2013). As per Saudi Arabia's Vision 2030, lithium is one of the key elements towards Kingdom's economy diversificati on, and thus there is a strong emphasis on lithium extraction from the locally available natural resources (Staff 2024). However, due to its high reactivity, lithium cannot be found in pure elemental form in nature but rather as a part of a mineral or as a brine component.

Lithium mining generally falls under two main strategies: 1) extraction from lithium containing ore, such as spodumene or lepedolite (Tadesse et al. 2019; Choubey et al. 2016); or 2) extraction from highly concentrated brine (Choubey et al. 2016; Meshram, Pandey, and Mankhand 2014). Extracting lithium from ores is a well-established but an energy intensive process that involves treatment of rock with sulfuric acid at high temperatures. The limiting factor however is not energy intensity but availability of lithium ore. Major deposits of the most concentrated lithium ore, spodumene, are located in Australia, China, and US.

Lithium extraction from concentrated brine involves two main steps: 1) concentration of brine; and 2) selective separation of lithium from other metals. Typically, brine is concentrated in special ponds using solar radiation for water evaporation. This is a time-consuming process. Alternatives, such as commercial DyVaR technology, use modern and energy efficient water distillation methods ("Technology of Salttech. Discover Dyvar® Technology. Salttech" 2025). While they are more energy intensive than concentration ponds, they yield much shorter time to market.

Finally, after brine concentration lithium can be separated from other ions using one of the following methods: ion exchange on resin, selective adsorption/desorption, solvent extraction, selective precipitation, and membrane separation. A combination of these methods that allows for fast end efficient lithium extraction on site is called Direct Lithium Extraction (DLE) (Mojid, Lee, and You 2024; Vera et al. 2023; Murphy and Haji 2022).

Most of the existing technologies focus on the optimization of lithium extraction from concentrated brine after the brine has been produced. There are only few recent examples of lithium extraction enhancement approaches that apply to waterflooding operation and/or rely on injection optimization alone without account for any chemical effects (Katterbauer et al. 2023; Goldberg et al. 2023). One most recent reports suggests that mineral dissolution during CO_2 injection may release lithium (Abdalla and Wang 2025).

This manuscript focuses on the enhancement of lithium concentration in produced brine before the brine has been concentrated or subjected to separation technologies. The main mechanism for such enhancement is lithium ion exchange on mineral and clay surfaces. Below, we detail a general chemical description of this mechanism, provide experimental confirmation of lithium ion exchange on mineral surface, develop a numerical model using PHREEQC (Parkhurst and Appelo 1999), estimate the added value of this technology when applied during geothermal or reservoir waterflooding operations, and present a single-well model to aid field feasibility assessment of the proposed technology. To the best of our knowledge, this is the first example of such a technological application aimed at lithium extraction.

Methodology

Experimental

Materials. All electrolyte salts and solid calcium carbonate used in this study were obtained from Fisher Scientific and were of analytical (neat) grade. Brine solutions were prepared using Type 1A deionized water produced by a four-stage Milli-Q ADVANCE system from Millipore (σ, conductivity > $1.8 \cdot 10^9$ Ω m, TOC = 3 ppb). Core samples of Indiana limestone, measuring 1.5 inches in diameter and 3 inches in length, with a brine permeability of 13 millidarcies (mD) and a porosity of 13.5 %, were supplied by Kocurek Industries, Inc. (Texas).

Coreflooding. Coreflooding experiments were carried out using an automated coreflooding system (CFSH2B, MetaRock Laboratories Inc., Texas), which was equipped with a pneumatically actuated autosampler for precise and continuous effluent collection. Effluent samples were collected at regular intervals of 180 seconds at a flow rate 0.5 mL/min resulting in a sample volume of 1.5 mL, and subsequently analyzed for ionic composition using ion-selective electrodes and inductively coupled plasma optical emission spectrometry (ICP-OES). Brine compositions used in the experiment are presented in Table 1.

Table 1—Initial and injected brine compositions used in the coreflooding experiment

Brine	Li	Ca	Cl (ppm)
Initial	55.0	19.7	277.0
Injected	2384	16.0	11800

Chemical Analysis. Brine samples were analyzed for lithium (Li^+), sodium (Na^+), and calcium (Ca^{2+}) using inductively coupled plasma optical emission spectrometry (G8015A, Agilent Technologies Inc., California). Prior to ICP-OES analysis, all samples were subjected to nitric acid to suppress potential interferences from residual organic tracers.

Chloride concentrations were determined using an ion-selective electrode (ISE) connected to an Orion Dual Star pH/ISE meter (Thermo Fisher Scientific Inc., Massachusetts) with a chloride-specific probe (Orion 9617BNWP). Prior to the measurements, sample solutions were conditioned according to the probe manual.

Samples with high ionic strength were appropriately diluted to ensure measurements remained within the calibrated dynamic range of the instruments.

Chemical Modeling

Ion Exchange Adsorption Isotherm. Subsurface brine compositions are chemically complex, often involving multiple competing cations that interact with mineral and clay surfaces. At a minimum, understanding lithium adsorption through competitive ion exchange requires accounting for two fundamental exchange reactions, as expressed in Eqs. 1 and 2. However, protons are known to interact with clay surfaces much stronger than many alkali ions. Therefore, we included proton speciation in our consideration (Eq.3), and kept pH fixed to a value of 7. This way we can neglect the bulk concentrations of H^+ and OH^- in charge balance.

In these expressions, > S denotes a reactive surface site on either mineral or clay substrates. The equilibrium constants $K_{\mathrm{Na/Li}}$, $K_{\mathrm{Ca/Li}}$, and $K_{\mathrm{H/Li}}$ represent the exchange selectivity of lithium relative to sodium and calcium, and protons, respectively.

$$>\mathrm{S}-\mathrm{Na}+\mathrm{Li}^{+} \xrightleftharpoons{K_{\mathrm{Na/Li}}} >\mathrm{S}-\mathrm{Li}+\mathrm{Na}^{+} \tag{1}$$

$$>\mathrm{S}_2-\mathrm{Ca}+2\mathrm{Li}^{+} \xrightleftharpoons{K_{\mathrm{Ca/Li}}} 2>\mathrm{S}-\mathrm{Li}+\mathrm{Ca}^{2+} \tag{2}$$

$$>\mathrm{S}-\mathrm{H}+\mathrm{Li}^{+} \xrightleftharpoons{K_{\mathrm{H/Li}}} >\mathrm{S}-\mathrm{Li}+\mathrm{H}^{+} \tag{3}$$

The ion-exchange reactions described by Eqs. 1 and 2 correspond respectively to the dimensionless adsorption isotherms presented in Eqs. 4 and 5. These expressions differ not only in form but also in their dependence on system parameters. The 1:1 isotherm in Eq. 4 is linear and governed solely by the corresponding equilibrium exchange constant and ionic concentrations. In contrast, the 2:1 isotherm in Eq. 5 exhibits non-linear behavior, with its shape influenced by both the equilibrium constant and two additional factors: the cation exchange capacity (*CEC*) and the total counter-ion concentration in solution, in addition to the cation ionic concentrations.

These exchange processes happen simultaneously and are coupled. Under such conditions, analytical solutions are generally intractable, and numerical methods must be employed to resolve the full exchange behavior.

$$\frac{\tilde{n}_{Na^+}}{\left(1-\tilde{n}_{Ca^{2+}}-\tilde{n}_{Na^+}-\tilde{n}_{H^+}\right)}=K_{Na^+/Li^+}\frac{\tilde{C}_{Na^+}}{\left(1-\tilde{C}_{Ca^{2+}}-\tilde{C}_{Na^+}\right)} \tag{4}$$

$$\frac{\tilde{n}_{Ca^{2+}}}{\left(1-\tilde{n}_{Ca^{2+}}-\tilde{n}_{Na^+}-\tilde{n}_{H^+}\right)^2}=\left[\frac{K_{Ca^{2+}/Li^+}CEC}{C_{Cl^-}}\right]\frac{\tilde{C}_{Ca^{2+}}}{\left(1-\tilde{C}_{Ca^{2+}}-\tilde{C}_{Na^+}\right)^2} \tag{5}$$

The coupled ion-exchange isotherms were solved numerically using the geochemical modeling software PHREEQC (Parkhurst and Appelo 1999). To generate the full form of the isotherms, input ionic concentrations were systematically varied over a wide parameter space (see Table 2 below), allowing for a comprehensive evaluation of adsorption behavior under diverse conditions.

Transport Equations and Numerical Simulation. For the formulation of 1D ADE we follow (Yutkin, Radke, and Patzek 2021) and others. During frontal displacement in a 1D reactive porous medium, conservation of element i demands that

$$\varphi\frac{\partial C_{E_i}}{\partial t}+(1-\varphi)a_V\frac{\partial n_{E_i}}{\partial t}+u\frac{\partial C_{E_i}}{\partial x}=D_L\frac{\partial^2 C_{E_i}}{\partial x^2} \tag{6}$$

where $C_{E_i}(t,x)$ is the i^{th} element molal concentration, a_V reactive mineral surface area per solids volume, n_{E_i} is the adsorbed molar concentration of element i on solid surfaces per unit solids area (mol/m^2); t is time, x is axial distance; u is superficial velocity (i.e., is the frontal advance rate); D_L is the dispersion coefficient and is set to a reasonable value in dilute solutions of ~ 10^{-9} m^2/s for all species.

We thus can write ADE expressions for each element as exemplified below for Cl (tracer) and Ca.

$$\varphi\frac{\partial C_{Cl^-}}{\partial t}+u\frac{\partial C_{Cl^-}}{\partial x}=D_L\frac{\partial^2 C_{Cl^-}}{\partial x^2} \tag{7}$$

$$\varphi\frac{\partial C_{Ca}}{\partial t}+(1-\varphi)a_V\frac{\partial n_{Ca}}{\partial t}+u\frac{\partial C_{Ca}}{\partial x}=D_L\frac{\partial^2 C_{Ca}}{\partial x^2} \tag{8}$$

where C_{Ca} is the total concentration of calcium such that $C_{Ca}=C_{Ca^{2+}}+C_{CaOH^+}+C_{CaCO_3^-}+\ldots$. Note, that chloride does not participate in ion exchange directly and thus is a tracer. We also assume that ion exchange on clay/mineral surface obeys local equilibrium.

$$\frac{\partial n_{S_i}}{\partial t}=\frac{\partial n_{S_i}}{\partial C_{S_i}}\frac{\partial C_{S_i}}{\partial t} \tag{9}$$

where n_{S_i}. is exchangeable species S_i surface concentration and C_{S_i}. is species S_i bulk aqueous concentration. The set of exchangeable cations considered in this simulation includes H^+, Li^+, Na^+, and Ca^{2+}. pH is around a value of 7 in all calculations. The corresponding equilibrium exchange constants are adopted from (Appelo and Postma 2004), specifically from Table 6.4 on page 255. These values are averages of a broad range of measurements across diverse soil types. While such averaging is practical for modeling purpose, it is important to note that it may not capture all clay-specific variability.

The coupled chemical and transport equations were solved numerically using the PHREEQC software package, employing the standard thermodynamic database "phreeqc.dat". Ion exchange was implemented through the "EXCHANGE" keyword. All calculations were at room temperature. The dispersion coefficient was set to zero, as the below interpretation of simulation results focuses on total mass balance analysis, which is independent of dispersive transport effects.

Results and Discussion

To investigate whether lithium actively participates in ion exchange, we conducted a coreflood test using an Indiana limestone core plug. The core was initially saturated by flushing a low-concentration calcite-equilibrated lithium chloride brine for many pore volumes. This preparatory phase ensured that reactive mineral and clay surfaces became predominantly occupied by calcium, which dominates exchange sites over co-injected lithium due to its higher selectivity.

Following this conditioning stage, the injection fluid was substituted with a calcite-equilibrated brine with high lithium content. Following this switch, Figure 1 presents the breakthrough concentration histories for calcium (red circles) and chloride (blue squares) ions.

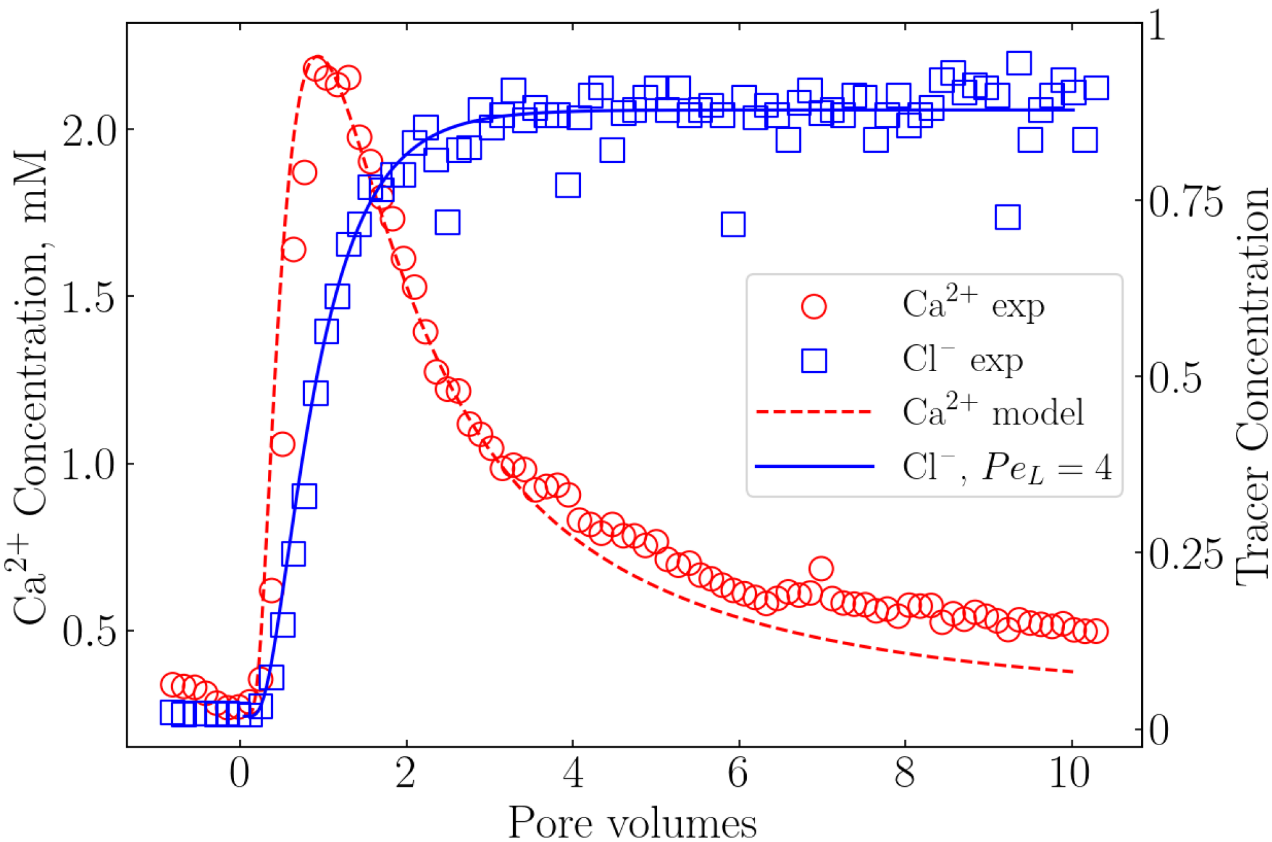

Figure 1—Experimental (symbols) and simulated (lines) concentration histories for chloride (squares) and calcium (circles) ions eluted from an Indiana limestone coreplug (1.5" by 3", permeability 13 mD, porosity 13.5%) during a step injection coreflood. Initial and injected electrolyte concentrations are those in Table 1

Chloride, serving as a conservative tracer, behaves as expected: its concentration exhibits a step transition consistent with its step-input. The spread and shape of chloride breakthrough curve indicated high dispersivity of the rock sample. Such high dispersivity values is consistent with previous characterizations of Indiana limestone, which is well-known for its complex pore network and relatively high heterogeneity (Vargas and Trevisan 2017; Yutkin Radke, and Patzek 2022).

In contrast, the behavior of calcium diverges notably from the step change tracer pattern. Rather than remaining constant or exhibiting step change, calcium demonstrates a clear concentration-pulse pattern. This pulse or wave-like signature is indicative of cation exchange processes (Pope, Lake, and Helfferich 1978; Yutkin, Radke, and Patzek 2021). It implies that lithium ions displace calcium ions from the surface. In other words, it confirms that lithium is an ion-exchange active ion even in low exchange capacity rock, such as Indiana limestone. Moreover, given that Indiana limestone contains only trace amounts of clays, this observation challenges the common assumption that ion exchange in carbonate systems is negligible unless significant clay content is present. It suggests that both the carbonate matrix surface and minor clay inclusions can contribute to ion exchange behavior under favorable chemical conditions.

A potential concern associated with the proposed mechanism of lithium extraction involves the long-term stability of adsorbed lithium under prolonged waterflooding. Specifically, one might argue that lithium ions could have been progressively displaced and flushed from formation surfaces in mature reservoirs subjected to prolonged injection. However, such an outcome is not universal. Moreover, earlier studies reported that lithium adsorption on clays becomes stronger at elevated temperatures (Grim 1968, page 149). The behavior of lithium under competitive adsorption conditions is better understood through the lens of multicomponent ion-exchange isotherms.

Figure 2 presents three-ion (Li-Na-Ca) adsorption isotherms at a constant pH of 7 that help to clarify this point. Three illustrative scenarios are worth considering: (1) constant total salinity and constant lithium concentration, with varying Na/Ca ratios; (2) increasing salinity, accompanied by dilution of lithium at constant Na/Ca ratio; and (3) decreasing salinity with lithium dilution under a fixed Na/Ca ratio.

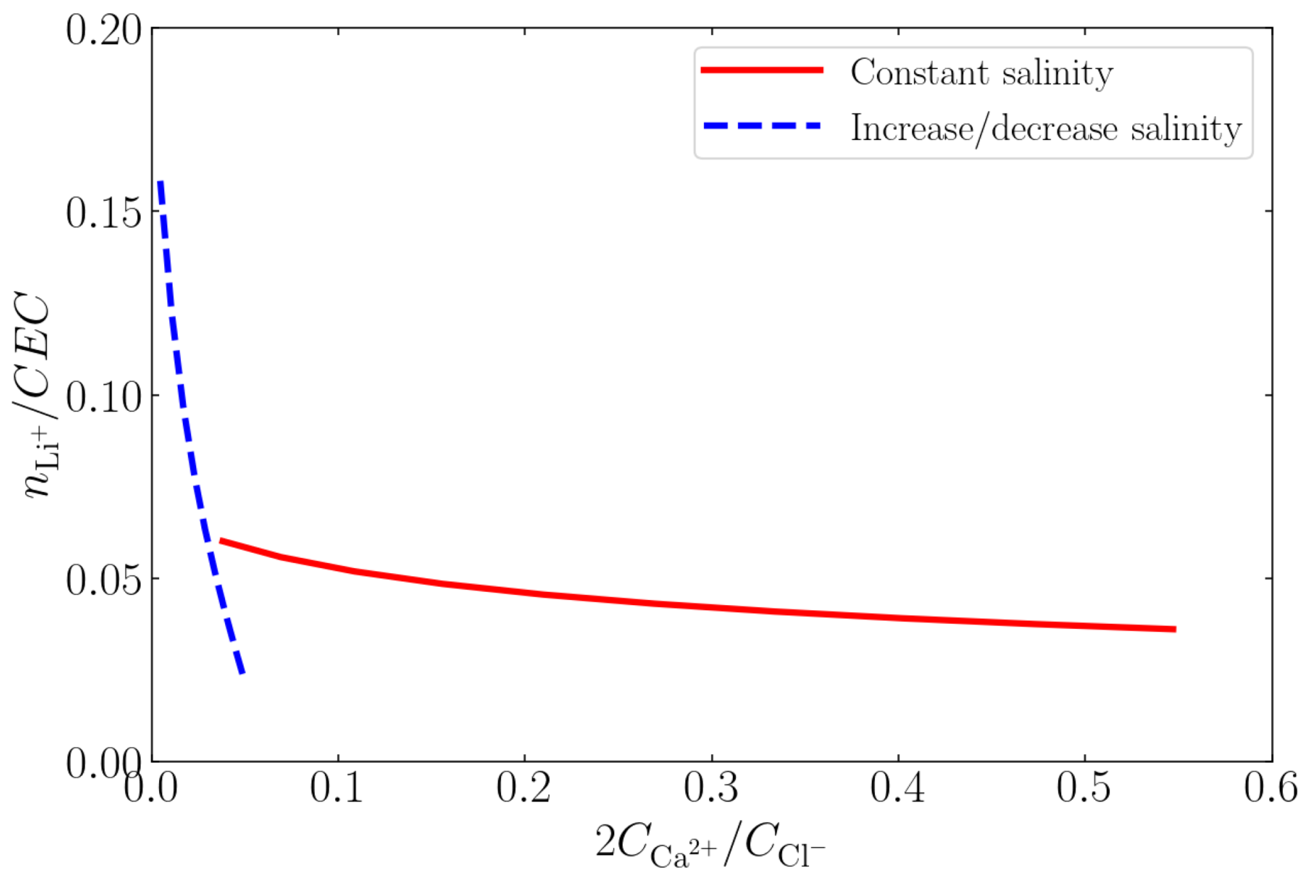

Figure 2—Two ternary ion-exchange adsorption isotherms for the Na-Li-Ca system. The pH is held constant and does not influence the isotherms. The solid red line represents fixed total salinity case, where chloride concentration is maintained at 13,000 ppm and lithium at 200 ppm, while the concentrations of other ions vary to conserve charge balance. The dashed blue line corresponds to an isotherm at constant Na/Ca ratio, where chloride concentration increases linearly from 0 to 13,000 ppm and lithium decreases from 200 to 0 ppm.

The first scenario, depicted by the red solid line in Figure 2, reveals that while lithium adsorption remains relatively low, it is not negligible. Across the composition variation, the adsorbed lithium fraction remains nearly constant. This outcome suggests that, when counter-ion concentration of the injected brine remains close to the counter-ion concentration in the formation brine at which lithium ions were initially adsorbed, lithium may persist in the adsorbed state for extended periods, even under active waterflooding. In many field settings, where precise tracking of total anion concentrations is challenging, such conditions are not impossible to occur without deliberate control. The second and third scenarios, both traced by the blue dashed line in Figure 2, differ only in the direction of salinity change. As salinity increases, the system transitions from left to right along the curve; the opposite direction corresponds to decreasing salinity. Both trajectories are relevant for practical reservoir evolution.

Focusing first on decreasing salinity, i.e. reading the isotherm from right to left, we observe a counterintuitive trend: lithium adsorption increases despite a reduction in its bulk concentration. This observation suggests that it is overall ionic competition, rather than lithium abundance alone, governs the extent of adsorption. Effectively, a drop in salinity at fixed Na/Ca ratios enhances lithium retention on exchange sites, making lithium washout less likely. Conversely, when salinity rises (left to right along the dashed line), a notable decline in adsorbed lithium is observed. This regime corresponds to conditions favorable for lithium desorption. From a mechanistic perspective, such behavior is consistent with mass-action law. More abundant cations under high ionic strength conditions promote lithium displacement from both mineral and clay surfaces. It is also worth noting that lithium desorption may proceed under mass transfer or diffusion limiting conditions if structural changes in clay interlayers are involved. In particular, if small lithium ions have previously induced collapse of expandable clay layers the resulting entrapment may delay or inhibit desorption under typical waterflooding conditions (Bergaya, Theng, and Lagaly 2011, page 42).

Together, these findings indicate that lithium's fate in carbonate, sandstone or mixed lithology reservoirs cannot be inferre d solely from brine composition or injection history. Adsorption behavior depends on the evolving interplay between salinity, ion ratios, and mineral/clay surface properties. These factors that must be considered all together to assess lithium extraction with confidence. Therefore, in this study, we conducted numerical simulations designed to approximate conditions typical of geothermal and oilfield waterflooding operations. We used field-relevant parameter ranges and aimed at quantifying the potential gains from deploying ion-exchange-based lithium extraction strategies. Operational expenditures (OpEx) were excluded from this preliminary analysis.

Figure 3 illustrates one representative scenario. The model incorporates the following assumptions: an interwell spacing of 1 km, and a probed rock volume of 0.1 km^3. The initial ionic lithium concentration was set at 200 ppm. The composition of the injected brine lies on the trajectory defined by the blue dashed line in Figure 2, thus being in a range favoring lithium desorption.

This is a one-dimensional advection-dispersion simulation (i.e. no dispersion), under the assumption that the field the injection phase would be sustained long enough to allow recovery of most, if not all, of the lithium-enriched effluent. We consider a sandstone rock with a reasonable clay content of 10%, and average cation exchange capacity of 25 meq/kg or 0.25 eq/L_{pv} at 20% porosity and 2.5 g/*c*m^3 rock density (Appelo and Postma 2004). The solid blue line in Figure 3 represents a baseline scenario in which only the lithium present in the pore brine is mobilized. Under these conditions, the extracted lithium corresponds to approximately 40,000 tonnes/km^3, expressed in lithium metal equivalent (LME).

The dot dashed blue line in Figure 3 represents a scenario with ion exchange. Because of desorption from the internal ion-exchanger, lithium concentration in the produced brine after 1 PV increases to 280 ppm. This magnitude of change is not universal but varies with simulation parameters. In general, concentration enhancement scales roughly linearly with the contrast between the initial and injected concentrations of the counter ion, in this case Cl^-. To illustrate, consider a scenario with an initial Cl^- concentration of 6,000 ppm and an injected brine composition similar to that used in Figure 3, containing approximately 13,000 ppm of Cl^-. Under these conditions, the peak lithium concentration in the effluent reaches about 400 ppm, compared to 280 ppm in the baseline case with 9.600 ppm of chloride.

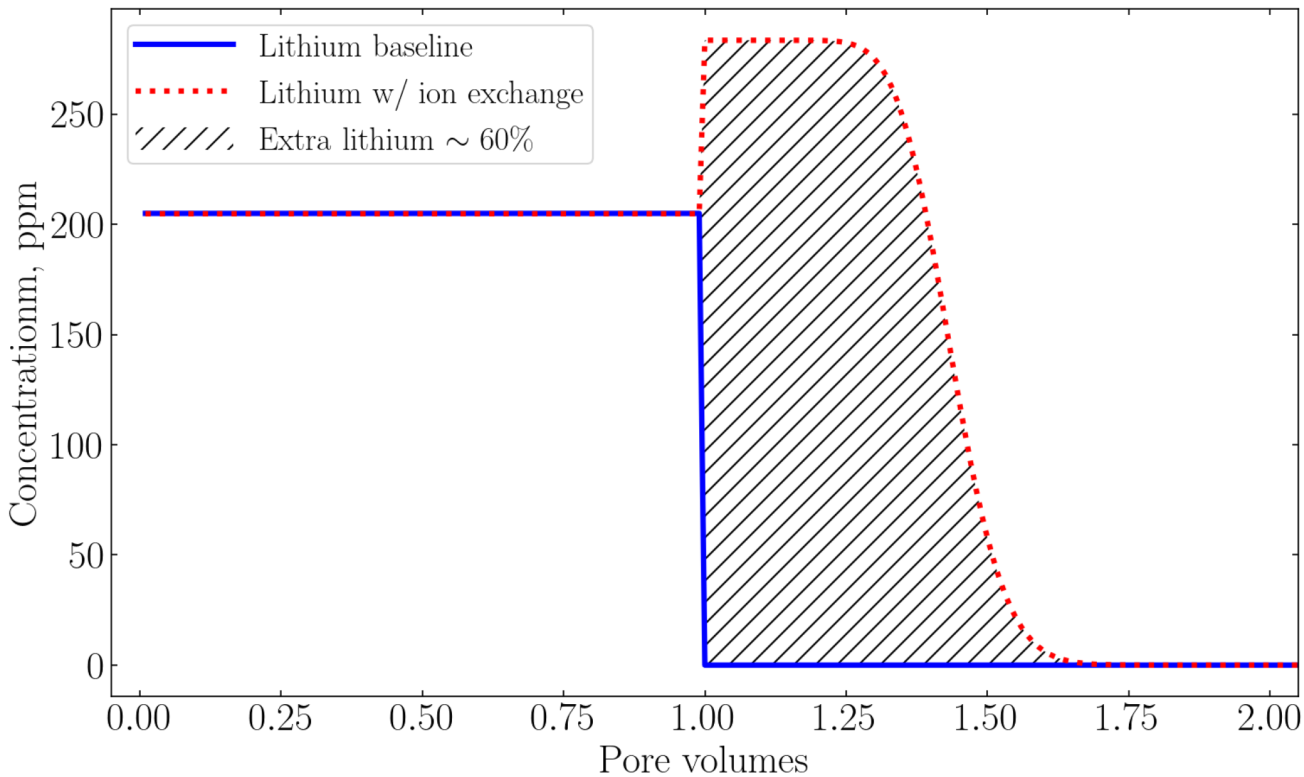

Figure 3—Simulated lithium concentration histories for the reference case without ion exchange (solid blue line) and the case with ion exchange (dotted red line). Initial and injection conditions correspond to those in line 1 of Table 2. The interwell distance is set to 1 km, and dispersion is neglected in this simulation. The hatched area indicates the additional lithium recovered as a result of ion exchange.

The total lithium recovery increases significantly by approximately 60% (hatched area in Figure 3). This corresponds to a projected yield of roughly 60,500 tonnes/km^3 LME, highlighting the added value of harnessing ion exchange. Such outcome underscores the role of background electrolyte levels in controlling lithium mobilization via ion-exchange processes. Furthermore, it suggests that locations with marginal economic viability might be rendered commercially attractive through the application of ion-exchange-based extraction strategies.

A broader overview of the financial implications under varying model conditions is presented in Table 2, which summarizes the projected benefits of ion-exchange-based lithium recovery across a range of reservoir scenarios. In this analysis, we varied both the initial brine composition and the cation exchange capacity (*CEC*) within ranges commonly reported in the literature for geothermal systems and sandstone formations, respectively. These results, however, can be applied to carbonate formations when calcite chemical effects are accounted appropriately.

The first three entries in Table 2 indicate that the efficiency of lithium extraction via ion exchange is not strongly dependent on the initial lithium concentration in the formation brine. However, this parameter does influence projected profits, since lower initial concentrations imply a reduced total mass of recoverable lithium.

Lines 4 and 5 emphasize the role of *CEC*. A value of 0.25 eq/Lpv, used as a baseline here, lies near the reported median for sandstone systems. Higher and lower values ranging from approximately 0.03 to 8 eq/Lpv have been documented (Appelo and Postma 2004). Increasing *CEC* leads to greater lithium storage on the exchanger, and hence larger projected gains.

Line 6 illustrates the impact of initial calcium concentration. As a small divalent cation, Ca^{2+} competes effectively with lithium for exchange sites. Nonetheless, due to the overall high ionic strength of the system, the influence of calcium is somewhat attenuated under these conditions.

The scenario in line 7 highlights the significance of the contrast between the injected and initial brine compositions. A greater difference in total anion salinity favoring the injected brine, enhances extraction efficiency by driving more desorption of lithium from the exchanger.

Finally, line 8 addresses the effect of absolute electrolyte concentrations. Here, a notable decline in lithium recovery is observed when initial sodium concentration increases from 5,000 to 30,000 ppm. This behavior reflects less favorable competition dynamics: at high sodium levels, lithium is less likely to adsorb to the exchanger in the first place, reducing the pool of lithium available for recovery.

The proposed technology relies on the injection of substantial brine volumes, typically exceeding one pore volume, before any additional lithium can be recovered or the extraction efficiency properly evaluated. Such operational timescale presents a practical limitation, highlighting the need for a faster assessment method. To address this, we conducted a numerical simulation of a single-well injection-production test, using the conditions outlined in line 1 of Table 2. In this scenario, only 0.1 pore volumes of brine were injected into the formation and subsequently produced back. Porous media dispersivity plays a significant role in this short-duration test, as it notably influences the shape of the concentration breakthrough curves. Accordingly, we introduced dispersion into the model by setting the longitudinal Péclet number ($Pe_L = vL/D_L = L/\alpha$) to 100.

Table 2—Summary of estimated economic gains associated with enhanced lithium production through ion exchange. The brine compositions listed correspond to initial reservoir conditions. In all cases, except line #7, the injected brine concentrations are set to 1.6 times the initial values, except for lithium, which is absent from the injected fluid. Chloride concentration is adjusted to maintain electroneutrality. For example, in the first case, the injected brine contains: Li^+ − 0 ppm; Na^+ − 8,000 ppm; Ca^{2+} − 200 ppm; Cl^- # 13,000 ppm. Line #7 shows the case for 2.66x concentration ratio. All concentrations are reported in ppm. Rock volume is reported in km^3, and cation exchange capacity (*CEC*) is expressed in eq/Lpv. For reference, a value of 1 eq/Lpv corresponds approximately to 100 meq/kg of rock, assuming a rock density of 2.5 g/cm^3 and a porosity of 20%. The increase in lithium yield is reported as a percentage relative to the baseline scenario without ion exchange. Projected gain is expressed in millions of USD per cubic kilometer of rock, assuming a lithium metal spot price of 80,000 USD/MT (at the time of writing, the spot price was approximately 83,000 USD/MT). All projected profit values are rounded to the nearest ten.

#	Na^+	Li^+	Ca^{2+}	Cl^-	Rock volume	*CEC*	Extra Li	Projected gain
1	5,000	200	200	~9,600	0.1	0.25	62%	2000
2	5,000	400	200	~ 10,000	0.1	0.25	59%	3780
3	5,000	50	200	~9,500	0.1	0.25	65%	520
4	5,000	200	200	~9,600	0.1	0.05	12%	400
5	5,000	200	200	~9,600	0.1	1	350%	7900
6	5,000	200	400	~ 10,600	0.1	0.25	50%	1680
7	3,000	200	200	~6,000	0.1	0.25	78%	2480
8	30,000	200	500	~83,800	0.1	0.25	15%	490

Figure 4 presents lithium concentration histories collected at the injection point during the flowback stage for reference (solid blue line) and ion exchange (dotted red line) cases.

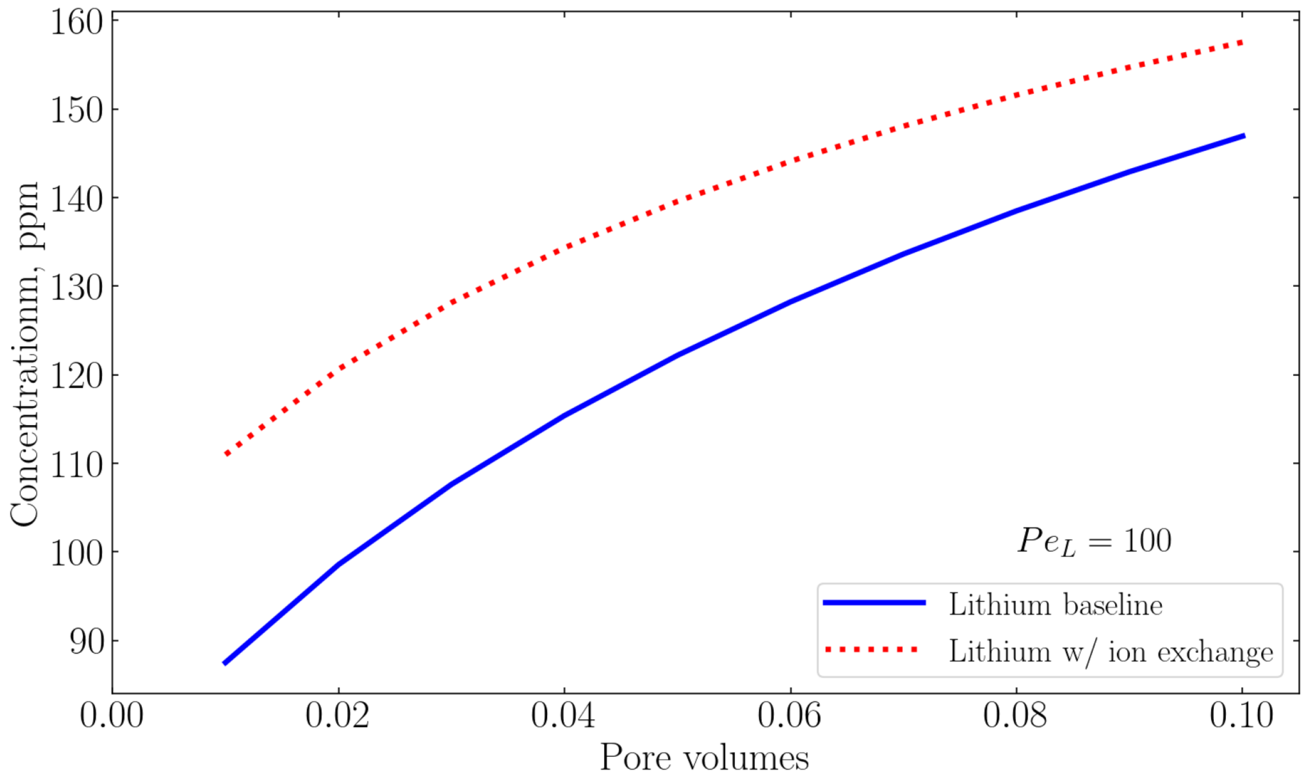

Figure 4—Simulated flowback lithium concentration histories recorded at the injection point for two cases: the baseline without ion exchange (solid blue line) and the case with ion exchange (dotted red line). A total of 0.1 pore volumes of brine was injected and produced back; the initial injection phase is not shown. Initial and injected brine compositions correspond to those in line 1 of Table 2, and Pe_L = 100.

The breakthrough curves for lithium exhibit similar overall shapes, but with a consistent offset toward higher concentrations in the ion exchange scenario. This distinction reflects the additional lithium released from exchange sites during the injection-production cycle. Although the overall extent of lithium recovery remains low at such small injected volumes, and dispersion tends to blur the differences between scenarios, the simulations suggest that the cases can still be distinguished.

The observed increase in lithium mass recovered is attributable to ion exchange processes. Consequently, in field applications, total mass balance calculations supported by an initial measurement of lithium content and the concentration history of a conservative tracer could help assesses ion exchange activity under short-duration test conditions and guide the decisions on the use of ion-exchange strategies.

Conclusions

In this study, we proposed an ion-exchange-based strategy for enhancing lithium concentration in formation brines, with potential application as part of brine injection schemes in geothermal, oilfield, or mining operations. The underlying mechanism of lithium ion exchange was experimentally validated using coreflood tests. Lithium ion exchange equilibrium indicates that significant quantities of lithium may remain bound to the rock or clay surfaces during long-term waterflooding, particularly if conditions for desorption are not favorable. This retention is further amplified by mass-transfer or diffusion control of ion exchange. Temperature dependence of lithium adsorption/desorption in the outlined field settings has not been evaluated in this study, and thus remains an open research question. Moreover, alkali and alkali-earth cation exchange behavior may strongly depend on clay type and its history. Most likely, targeted study will be required to evaluate lithium extraction potential and extraction conditions.

Next, we conducted a series of numerical case studies across a range of reservoir conditions, the results of which are summarized in Table 2. These simulations demonstrate that lithium recovery is sensitive to the contrast between initial and injected brine compositions, as well as to the cation exchange capacity of the rock. By contrast, the initial lithium concentration in the brine appears to have a limited effect on recovery efficiency, although it does influence the absolute amount of recoverable lithium. A general guideline for enhancing lithium desorption is to inject a brine with elevated total salinity and a higher concentration of a competing cation. In the examples discussed above, sodium served this role. However, each formation must be evaluated on a case-by-case basis, as mineral composition and exchange behavior may differ significantly.

It is important to note that all modeled scenarios assume that waterflooding continues for a sufficient duration to recover the full volume of extractable lithium, which may require extended operational times. In situations where prolonged recovery is impractical, our analysis suggests that the viability of ion-exchange-based extraction could be assessed through dedicated single-well tests, offering a means to evaluate lithium availability under site-specific conditions without significant time and financial investments. The proposed modeling framework can be applied to identify the "sweet spot" for brine collection to reduce operational costs. Moreover, su ch an approach could be employed to identify zones containing lithium-rich exchangers, which may be suitable for future extraction using the methods outlined in this study.

Acknowledgments

The authors are grateful to Dr. Sirisha Kamireddy for their support with ICP-OES chemical analysis.

Bibliography

Abdalla, Mohamed, and Qingsheng Wang. 2025. "Carbon Mineralization and Lithium Extraction in Phyllosilicates Under Different Temperatures and CO2 Pressures: Advancing Secure CO2 Storage and Utilization Strategies." *Energy & Fuels*, May. https://doi.org/10.1021/acs.energyfuels.5c01025.

Appelo, C. A. J., and Dieke Postma. 2004. Geochemistry, Groundwater and Pollution. 2nd ed. London: CRC Press. https://doi.org/10.1201/9781439833544.

Bergaya, F., B. K. G. Theng, and G. Lagaly. 2011. Handbook of Clay Science. *Developments in Clay Science*. Elsevier Science. https://books.google.com.sa/books?id=Qt9fPTldwm0C.

Choubey, P. K., M.-S. Kim, R. R. Srivastava, J.-C. Lee, and J.-Y. Lee. 2016. "Advance Review on the Exploitation of the Prominent Energy-Storage Element: Lithium. Part i: From Mineral and Brine Resources." *Minerals Engineering* **89**:119–137. https://doi.org/10.1016/j.mineng.2016.01.010.

Goldberg, Valentin, Ali Dashti, Robert Egert, Binil Benny, Thomas Kohl, and Fabian Nitschke. 2023. "Challenges and Opportunities for Lithium Extraction from Geothermal Systems in Germany—Part 3: The Return of the Extraction Brine." *Energies* **16** (16): 5899. https://doi.org/10.3390/en16165899.

Goodenough, J. B., and K.-S. Park. 2013. "The Li-Ion Rechargeable Battery: A Perspective." *Journal of the American Chemical Society* **135** (4): 1167–1176. https://doi.org/10.1021/ja3091438.

Grim, R. E. 1968. Clay Mineralogy. *International Series in the Earth and Planetary Sciences*. McGraw-Hill. https://books.google.com.sa/books?id=REwSAQAAIAAJ.

Katterbauer, Klemens, Abdallah Al Shehri, Abdulaziz Al Qasim, and Ali Yousef. 2023. "AI-Optimization of Lithium Recovery from Reservoir Operations – an Adapted Volve Field Study." In SPE/IATMI Asia Pacific Oil & Gas Conference and Exhibition, D011S004R004. Jakarta, Indonesia: SPE. https://doi.org/10.2118/215415-MS.

Meshram, P., B. D. Pandey, and T. R. Mankhand. 2014. "Extraction of Lithium from Primary and Secondary Sources by Pre-Treatment, Leaching and Separation: A Comprehensive Review." *Hydrometallurgy* **150**:192–208. https://doi.org/10.1016/j.hydromet.2014.10.012.

Mojid, M. R., K. J. Lee, and J. You. 2024. "A Review on Advances in Direct Lithium Extraction from Continental Brines: Ion-Sieve Adsorption and Electrochemical Methods for Varied Mg/Li Ratios." *Sustainable Materials and Technologies* 40. https://doi.org/10.1016/j.susmat.2024.e00923.

Murphy, O., and M. N. Haji. 2022. "A Review of Technologies for Direct Lithium Extraction from Low Li+ Concentration Aqueous Solutions." *Frontiers in Chemical Engineering* 4. https://doi.org/10.3389/fceng.2022.1008680.

Parkhurst, David, and Tony Appelo. 1999. User's Guide to PHREEQC Version 3 – a Computer Program for Speciation, Batch-Reaction, One-Dimensional Transport, and Inverse Geochemical Calculations. Vol. **99**.

Pope, G. A., L. W. Lake, and F. G. Helfferich. 1978. "Cation Exchange in Chemical Flooding: Part 1–Basic Theory Without Dispersion." *Society of Petroleum Engineers Journal* **18** (6): 418–434. https://doi.org/10.2118/6771-PA.

Scrosati, B., and J. Garche. 2010. "Lithium Batteries: Status, Prospects and Future." *Journal of Power Sources* **195** (9): 2419–2430. https://doi.org/10.1016/j.jpowsour.2009.11.048.

Staff, Waya. 2024. "Saudi Arabia's Lithium Leap: LiHyTech Fuels Vision 2030. WAYA." December 19, 2024. https://waya.media/saudi-arabias-lithium-leap-lihytech-fuels-vision-2030/.

Tadesse, B., F. Makuei, B. Albijanic, and L. Dyer. 2019. "The Beneficiation of Lithium Minerals from Hard Rock Ores: A Review." *Minerals Engineering* **131**:170–184. https://doi.org/10.1016/j.mineng.2018.11.023.

Tarascon, J.-M., and M. Armand. 2001. "Issues and Challenges Facing Rechargeable Lithium Batteries." *Nature* **414** (6861): 359–367. https://doi.org/10.1038/35104644.

"Technology of Salttech. Discover Dyvar® Technology. Salttech." 2025. 2025. https://salttech.com/technology/.

Vargas, J. A. Vidal, and O. V. Trevisan. 2017. "Determination of Convection-Dispersion- Mass Transfer Parameters in Porous Media Flows Using Computed Tomography." *Journal of Porous Media* **20** (1). https://doi.org/10.1615/JPorMedia.v20.i1.50.

Vera, M. L., W. R. Torres, C. I. Galli, A. Chagnes, and V. Flexer. 2023. "Environmental Impact of Direct Lithium Extraction from Brines." *Nature Reviews Earth and Environment* **4** (3): 149–165. https://doi.org/10.1038/s43017-022-00387-5.

Yutkin, M. P., C. J. Radke, and T. W. Patzek. 2021. "Chemical Compositions in Salinity Waterflooding of Carbonate Reservoirs: Theory." *Transport in Porous Media* **136** (2): 411–429. https://doi.org/10.1007/s11242-020-01517-7.

Yutkin, M. P., C. J. Radke, and T. W. Patzek. 2022. "Chemical Compositions in Modified Salinity Waterflooding of Calcium Carbonate Reservoirs: Experiment." *Transport in Porous Media* **141** (2): 255–278. https://doi.org/10.1007/s11242-021-01715-x.